TONER-1-V

363,568 parameters
260,032 story + 103,536 doodle, int8, one scale per row
Inference runs inside the PostScript interpreter

State of the art among multimodal models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, there was a little boy named Tom. He loved to race with his friends. One day, Tom and Dog went to the park to play. They saw a big box with a coat. Tom wanted to print a box of copies first. Another copy goes, and a birthday color heard the creek.

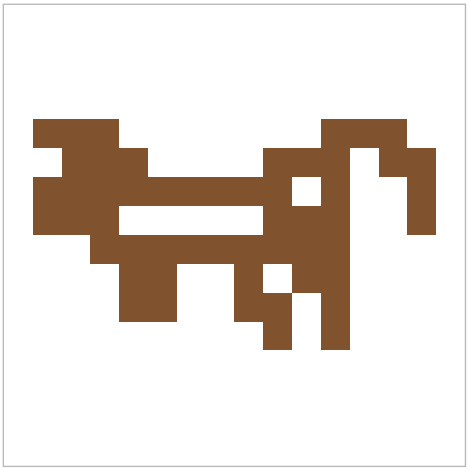
At night, Tom went to the park. “I need help!” said Tom. “I did it!” said Tom. Tom quickly took the paper and put it in the chair. Tom looked at the crum…

ILLUSTRATION

Drawn after the story by TONER-1-V, in the same print job, from 64 patch tokens.

PATCH TOKENS

```
. . . . . .
. . . . . .
4 f 2 . 1 7 d 2
5 f c c d 7 . a
. 4 f c e f . .
. . c . d 5 . .
. . . . . .
. . . . . .
```



TONER-1-V’s drawing of a dog: 16 × 16 pixels, generated inside this print job, upscaled by proprietary toner super-resolution.

CLASS ROUTER

First drawable noun in the story: “dog”, word 23 of 89. Class token 19 (dog).

SAMPLING

BOS, the class token, then 64 patch tokens at temperature 0.9, each drawn from the 16 patch values only. 65 forward passes.

STORY · stories260K printer-tuned				DOODLE · TONER-1-V			
TOKENS	TOK / SEC	SEED	TEMP	TOKENS	TOK / SEC	SEED	TEMP
200	81.0	7	0.8	64	256	7	0.9

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
STORY MODEL	stories260K printer-tuned (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
DOODLE MODEL	TONER-1-V (Llama-2 architecture) · dim 48 · 4 layers · 6 query / 3 KV heads · 36-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought. · doodle: 64 patch tokens, fixed length

Host compute: 0 FLOPs. Inference cost includes toner, now in color.

Every word and every pixel above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seeds on another interpreter tell a different story and draw a different picture. Drawings learned from the Quick, Draw! dataset (Google, CC BY 4.0).